Solar Collector Array: System Design Description

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Revision	B
Status	APPROVED
Classification	INTERNAL
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1. System overview

The solar collector array supplements the reactor with photovoltaic panels on tracking drives around the station's sunward hemisphere.

The system is led by Reactor & Power Systems. It is delivered through 5 work packages with a combined budget of 52,528.26 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
SA-SCP	Solar collector panel	SHV-SA-SPEC-0101	SHV-SA-DWG-0201	SHV-SA-BOM-0301
SA-SAT	Array tracking drive	SHV-SA-SPEC-0102	SHV-SA-DWG-0202	SHV-SA-BOM-0302
SA-PCU	Power conditioning unit	SHV-SA-SPEC-0103	SHV-SA-DWG-0203	SHV-SA-BOM-0303

3. Architecture

3.1 Solar collector panel

The solar collector panel is specified in SHV-SA-SPEC-0101. Its governing parameter is the panel area, set at 749 m². The responsible engineer is Lannic Crestwell.

3.2 Array tracking drive

The array tracking drive is specified in SHV-SA-SPEC-0102. Its governing parameter is the pointing accuracy, set at 0.42 deg. The responsible engineer is Nadia Mathers.

3.3 Power conditioning unit

The power conditioning unit is specified in SHV-SA-SPEC-0103. Its governing parameter is the rated power, set at 1.8 GW. The responsible engineer is Orrin Elsmore.

4. Interfaces

Document	Title
SHV-ICD-SA-PD	Interface Control Document: Solar Collector Array / Power Distribution

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-SA-01	Solar Collector Array: Design & analysis	Alira Aster	IN_PROGRESS	78.3%

Work package	Name	Owner	Status	Complete
WP-SHV-SA-02	Solar Collector Array: Fabricate & install solar collector panel	Jessa Pell	IN_PROGRESS	88.0%
WP-SHV-SA-03	Solar Collector Array: Fabricate & install array tracking drive	Edric Welles	IN_PROGRESS	92.7%
WP-SHV-SA-04	Solar Collector Array: Fabricate & install power conditioning unit	Alira Aster	IN_PROGRESS	87.3%
WP-SHV-SA-05	Solar Collector Array: Integration & test	Esme Sable	IN_PROGRESS	91.3%

6. Risks

- Supply of solar collector panel assemblies depends on a single supplier.
- Supply of array tracking drive assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.